

Validity test for all centralities in the code

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September 2024

CONTENTS

1	Introduction	2
2	Centralities	2
2.1	Degree centrality	2
2.2	Closeness centrality	3
2.3	Betweenness centrality	3
2.4	Eigenvector centrality	4
2.5	Bonacich centrality	5
2.6	König's paper centrality (Katz-Bonacich)	6

1 INTRODUCTION

Before doing the test, let's create a dummy adjacency matrices with dimensions 5x5. One for enmity and one for friendship.

▲	A	B	C	D	E
A	0	1	0	1	1
B	1	0	1	1	0
C	0	1	0	1	0
D	1	1	1	0	0
E	1	0	0	0	0

Figure 1: Adjacency matrix (enmity)

▲	A	B	C	D	E
A	0	0	1	0	0
B	0	0	0	0	1
C	1	0	0	0	1
D	0	0	0	0	1
E	0	1	1	1	0

Figure 2: Adjacency matrix (enmity)

2 CENTRALITIES

2.1 Degree centrality

Formula: $\frac{d_i(g)}{n-1}$

▲	Degree centrality	▼
A	0.75	
B	0.75	
C	0.50	
D	0.75	
E	0.25	

Figure 3: Degree centrality results

2.2 Closeness centrality

Formula: $\frac{(n-1)}{\sum_{i \neq j} l(i,j)}$

Closeness centrality	▼
0.8000000	
0.8000000	
0.5714286	
0.8000000	
0.5000000	

Figure 4: Closeness centrality results

2.3 Betweenness centrality

Formula: $\sum_{k \neq j, i \notin \{k,j\}} \frac{P_i(kj)/P(kj)}{(n-1)(n-2)/2}$

Betweenness centrality	
	0.5000000
	0.1666667
	0.0000000
	0.1666667
	0.0000000

Figure 5: Betweenness centrality results

2.4 Eigenvector centrality

Calculation of Eigenvector centrality uses power iteration method.

1. Take adjacency matrix
2. Multiply with all-ones vector
3. Normalize the resultant vector by dividing each element by the square root of the sum of the squares
4. Multiply adjacency matrix with new vector
5. Follow steps 3-4 for 12 iterations

Eigenvector centrality
0.4746025
0.5371662
0.4065931
0.5371662
0.1798316

Figure 6: Eigenvector centrality results

2.5 Bonacich centrality

Solving the following equation yield the Bonacich centrality for each agent

$$(\mathbf{I} - \beta A)^{-1} \cdot \mathbf{1}$$

Bonacich centrality
-3.00
-3.75
-2.75
-3.75
-0.50

Figure 7: Bonacich centrality results

2.6 Konig's paper centrality (Katz-Bonacich)

Katz–Bonacich centrality 	
	1.787429
	3.462154
	2.221360
	3.462154
	–1.797979

Figure 8: Katz-Bonacich centrality